Network analysis Branin Feb 2, 1959

TCTAP Program. Transient analysis.

- computed circuits with transistors predicted rise times within 0.1 mμs.
  current mode switching circuits.
  non-saturating circuits.

up to 20 transistors. (8 transistors 35-45 minutes on 704.

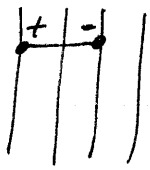
Need :- algebraic & differential eqs.
- Solve equations.

Kron's Method

upto 200 branches, 100 node pairs, 100 mesh points in 2 or 3 minutes.

Matrix Method
- Ohm's law
- wheatstone bridge

connections



connections between nodes
+1 −1

node wires

Branch node incidence matrix.

"Tree branches" & "links" —

Kirchoff voltage law & current law.

Voltage law

$e = Ae'$

or $C_t e = 0$

Current law

$i = Ci'$

$A_t i = 0$

Mesh Method

$E + e = Z(I + i)$

$(E - ZI) + e = Zi$

premply: $C_t(E - ZI) = C_t Z i$

$C_t(E - ZI) = C_t Z C i'$

$i' = (C_t Z C)^{-1} C_t (E - ZI)$

$i = C(C_t Z C)^{-1} C_t (E - ZI)$

$e = [Z - ZC(C_t Z C)^{-1} C_t Z](I - YE)$

$(C_t Z C)^{-1}$ Mesh Soln. Matrix

Node Method

$I + i = Y(E + e)$

$(A_t Y A)^{-1}$ Nodal Soln. Matrix

Kron's method

tree links $\begin{bmatrix} e_T \\ e_L \end{bmatrix} = \begin{bmatrix} A_T \\ A_L \end{bmatrix} [e']$ where $e_T = A_T e'$

$e' = (A_T)^{-1} e_T = (B_T)_t e_T$

where $(B_T)_t \equiv (A_T)^{-1}$

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generate network block at a time.

Some times for matrix inversion

$P = ab$ $\qquad 0.2\,M P^2 = 0.2 a^3 b^3$

$M = ab$ $\qquad 0.2\,P^3$ milliseconds for usual method

if all diffat $0.4 a^2 b^3$ } full soln

if all alike $0.267 a^2 b^3$ }

if	a	b	P = node pairs	complete soln $.2a^2b^3$	all diff $0.4a^2b^3$	all alike $0.267ab$	partial soln	
dimensions of network	31	32	992	55 hrs.	3.5 hrs	2.3	33 min	2.2 min
	63	64	4032	3580 hrs	113 hrs.	76	9 hrs.	1.0
	127	128	16,156	234,000 hrs	3680 hrs.	2460 hrs.	145 h.	12.7

on 704

Numerical Integration — biggest bottleneck.